

1) Student Information

</java>

```
class student {
```

```
    private String name;
```

```
    private int rollNo;
```

```
    private String faculty;
```

```
    public student(String name, int rollNo, String faculty) {
```

```
        this.name = name;
```

```
        this.rollNo = rollNo;
```

```
        this.faculty = faculty;
```

```
    }
```

```
    public String getName() {
```

```
        return name;
```

```
    }
```

```
    public int getRollNo() {
```

```
        return rollNo;
```

```
    }
```

```
    public String getFaculty() {
```

```
        return faculty;
```

```
    }
```

```
    public void setName(String name) {
```

```
        this.name = name;
```

```
    }
```

```
    public void setRollNo(int rollNo) {
```

```
        this.rollNo = rollNo;
```

```
    }
```

```
public void set faculty (string faculty) {  
    this.faculty = faculty;  
}
```

```
public void display profile() {  
    system.out.println("name:" + name);  
    system.out.println("Roll No:" + rollNo);  
    system.out.println("faculty:" + Faculty);  
}
```

```
}
```

```
public class main {
```

```
    public static void main (String[] args) {
```

```
        student s = new student("Aayushma", 1, "BCSIT");
```

```
        s.display profile();
```

```
    }
```

```
}
```


Project

main.java

C:\Users\acer\IdeaProjects\main.java

> .idea

> out

> src

> Main.java

.gitignore

main.java

main.java.iml

External Libraries

Scratches and Consoles

Main.java

Trial

1 class Student { 2 usages

32 public void setFaculty(String faculty) { no usages

33 | this.faculty = faculty;

34 | }

35

36 public void displayProfile() { 1 usage

37 | System.out.println("Name: " + name);

38 | System.out.println("Roll No: " + rollNo);

39 | System.out.println("Faculty: " + faculty);

40 | }

41 | }

42

43 public class Main {

44 | public static void main(String[] args) {

45 | | Student s = new Student(name: "Aayushma", rollNo: 1, faculty: "BCSIT");

46 | | s.displayProfile();

Run

Main

C:\Users\acer\.jdk\openjdk-26.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\In

Name: Aayushma

Roll No: 1

Faculty: BCSIT

Process finished with exit code 0

Performance

Show Results

00:00

Process Terminated

main.java > src > Main.java

48:2 LF UTF-8 4 spaces

2) Employee inheritance

// java

```
class Employee {  
    void displayDetails() {  
        System.out.println("Employee");  
    }  
}
```

```
class Teacher extends Employee {  
    void displayDetails() {  
        System.out.println("Teacher");  
    }  
}
```

```
class AdminStaff extends Employee {  
    void displayDetails() {  
        System.out.println("Admin Staff");  
    }  
}
```

```
public class main {  
    public static void main (String [] args) {  
        Employee t = new Teacher ();  
        Employee a = new AdminStaff ();  
    }  
}
```

```
    t.displayDetails();
```

```
    a.displayDetails();  
}
```

```
}
```

main.java

Version Control

Current File

Trial

Project

main.java

C:\Users\acer\IdeaProjects\main.java

.idea

out

src

Main.java

.gitignore

main.java

main.java.iml

External Libraries

Scratches and Consoles

Main.java

Trial

```
13
14 class AdminStaff extends Employee { 1 usage
15     @Override 2 usages
16     void displayDetails() {
17         System.out.println("Admin Staff");
18     }
19 }
20
21 public class Main {
22     public static void main(String[] args) {
23         Employee t = new Teacher();
24         Employee a = new AdminStaff();
25
26         t.displayDetails();
27         a.displayDetails();
28     }
29 }
```

Run

Main

C:\Users\acer\.jdk\openjdk-26.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\Intell:
Teacher
Admin Staff
Process finished with exit code 0

Performance

Show Results

00:00

Process Terminated

main.java > src > Main.java

29:2 LF UTF-8 4 spaces

24°C Mostly cloudy

Search

9:09 AM 6/22/2026

3) Course Registration

[/java]

```
import java.util. ArrayList;
```

```
class course {
```

```
    String courseName;
```

```
    course (String courseName) {
```

```
        this.courseName = courseName;
```

```
    }
```

```
}
```

```
class student {
```

```
    String name;
```

```
    ArrayList<course> courses = new ArrayList<>();
```

```
    student (String name) {
```

```
        this.name = name;
```

```
    }
```

```
    void addcourse (course c) {
```

```
        courses.add(c);
```

```
    }
```

```
    void displaycourses() {
```

```
        System.out.println("courses:");
```

```
        for (course c : courses) {
```

```
            System.out.println(c.courseName);
```

```
        }
```

```
    }
```

```
}
```

```
public class main {  
    public static void main(String[] args) {  
        student s = new student("Ram");  
  
        s.addcourse(new course("java"));  
        s.addcourse(new course("database"));  
  
        s.displaycourses();  
    }  
}
```


4.) Abstract student fee

. // java

```
abstract class student {
```

```
    abstract double calculate fee();
```

```
}
```

```
class undergraduate extends student {
```

```
    double calculate fee() {
```

```
        return 50000;
```

```
    }
```

```
}
```

```
class Graduate extends student {
```

```
    double calculate fee() {
```

```
        return 80000;
```

```
    }
```

```
}
```

```
public class main {
```

```
    public static void main(String[] args) {
```

```
        student u = new undergraduate();
```

```
        student g = new Graduate();
```

```
        System.out.println("UG fee=" + u.calculate fee());
```

```
        System.out.println("Graduate fee=" + g.calculate  
fee());
```

```
    }
```

```
}
```

The image shows a screenshot of an IDE (likely IntelliJ IDEA) with a dark theme. The top bar shows the 'main.java' file and 'Version Control' tabs. The left sidebar displays the project structure: 'main.java' (C:\Users\acer\IdeaProjects\main.java) with subfolders '.idea', 'out', and 'src'. The 'src' folder contains 'Main.java', '.gitignore', 'main.java', and 'main.java.iml'. The main editor window shows the 'Main.java' file with the following code:

```
9 }  
10  
11 class Graduate extends Student { 1 usage  
12     double calculateFee() { 2 usages  
13         return 80000;  
14     }  
15 }  
16  
17 public class Main {  
18     public static void main(String[] args) {  
19         Student u = new Undergraduate();  
20         Student g = new Graduate();  
21  
22         System.out.println("UG Fee = " + u.calculateFee());  
23         System.out.println("Graduate Fee = " + g.calculateFee());  
24     }  
25 }
```

The bottom panel shows the 'Run' tab with the command: `C:\Users\acer\.jdk\openjdk-26.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ\bin\idea_rt.jar=26.0.1:C:\Program Files\JetBrains\IntelliJ\bin\idea.jar" -jar C:\Users\acer\IdeaProjects\main\out\production\main\Main.jar`. The output shows: `UG Fee = 50000.0` and `Graduate Fee = 80000.0`. The process finished with exit code 0. The right side of the bottom panel shows 'Performance' and 'Process Terminated'.

5.) Interface Result processor

// Java

```
interface ResultProcessor {  
    void calculateGrade(int marks);  
}
```

```
class EngineeringDepartment implements ResultProcessor {  
    public void calculateGrade(int marks) {  
        System.out.println(marks >= 80 ? "Grade A" : "Grade B");  
    }  
}
```

```
class ManagementDepartment implements ResultProcessor {  
    public void calculateGrade(int marks) {  
        System.out.println(marks >= 70 ? "Grade A" : "Grade B");  
    }  
}
```

```
public class Main {  
    public static void main (String[] args) {  
        ResultProcessor e = new EngineeringDepartment();  
        ResultProcessor m = new ManagementDepartment();  
  
        e.calculateGrade(85);  
        m.calculateGrade(75);  
    }  
}
```

main.java

Version Control

Current File

▶

⚙️

⋮

🌐

🔍

⚙️

Trial

—

📄

✕

Project

main.java

C:\Users\acer\IdeaProjects\main.java

.idea

out

src

Main.java

.gitignore

main.java

main.java.iml

External Libraries

Scratches and Consoles

Main.java

Trial

class EngineeringDepartment implements ResultProcessor {

10

}

11

class ManagementDepartment implements ResultProcessor { 1 usage

12

public void calculateGrade(int marks) { 2 usages

13

System.out.println(marks >= 70 ? "Grade A" : "Grade B");

14

}

15

}

16

public class Main {

17

public static void main(String[] args) {

18

ResultProcessor e = new EngineeringDepartment();

19

ResultProcessor m = new ManagementDepartment();

20

21

e.calculateGrade(marks: 85);

22

m.calculateGrade(marks: 75);

23

}

24

}

25

Run

Main

▶

🔄

📷

🔗

⋮

↑

C:\Users\acer\.jdk\openjdk-26.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA\bin\idea_rt.jar=12121:C:\Users\acer\AppData\Local\Temp\jetbrains-idea-runtime-26.0.1\bin\java.exe"

↓

Grade A

Grade A

Process finished with exit code 0

Performance

Show Results

00:00

Process Terminated

main.java

src

Main.java

25:2

LF

UTF-8

4 spaces

24°C

Mostly cloudy

Search

🌳

🖨️

🔄

📁

🌐

📅

🔍

🔗

🔊

🔋

9:11 AM

6/22/2026

🔔

6) library members

// java

```
class member {
```

```
    int borrowingLimit()
```

```
    { return 0;
```

```
    }
```

```
}
```

```
class studentMember extends member {
```

```
    @Override
```

```
    int borrowingLimit() {
```

```
        return 3;
```

```
    }
```

```
}
```

```
class TeacherMember extends member {
```

```
    @Override
```

```
    int borrowingLimit() {
```

```
        return 10;
```

```
    }
```

```
}
```

```
public class main {
```

```
    public static void main(String[] args) {
```

```
        System.out.println("student limit: " + new studentMember().borrowingLimit());
```

```
        System.out.println("Teacher limit: " + new TeacherMember().borrowingLimit());
```

```
    }
```

```
}
```


main.java

Version Control

Current File

Trial

Project

main.java

out

src

Main.java

.gitignore

main.java

main.java.iml

External Libraries

Scratches and Consoles

Main.java

Trial

```
13
14 class TeacherMember extends Member { 1 usage
15     @Override 2 usages
16     int borrowingLimit() {
17         return 10;
18     }
19 }
20
21 public class Main {
22     public static void main(String[] args) {
23         System.out.println("Student Limit: " +
24             new StudentMember().borrowingLimit());
25
26         System.out.println("Teacher Limit: " +
27             new TeacherMember().borrowingLimit());
28     }
29 }
```

Run

Main

C:\Users\acer\.jdk\openjdk-26.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA\lib\idea_rt.jar=12121:C:\Program Files\JetBrains\IntelliJ IDEA\bin\idea.exe" -Didea.config.path=C:\Users\acer\AppData\Local\JetBrains\IntelliJ IDEA\config\idea.config.xml -Didea.platform.prefix=Java -Didea.version=2025.1.1 -Didea.vendor.id=JetBrains -Didea.platform.prefix=Java -Didea.version=2025.1.1 -Didea.vendor.id=JetBrains

Student Limit: 3
Teacher Limit: 10
Process finished with exit code 0

Performance
Show Results
00:00
Process Terminated

main.java > src > Main.java

24°C
Mostly cloudy

Search

Taskbar icons: File Explorer, Edge, VS Code, IntelliJ IDEA, Chrome

System tray: 9:12 AM 6/22/2026

7.) Attendance

// java

```
class Attendance {  
    double calculateAttendance(int attended, int total) {  
        return 0;  
    }  
}
```

```
class EngineeringAttendance extends Attendance {  
    @Override  
    double calculateAttendance(int attended, int total) {  
        return attended * 100.0 / total;  
    }  
}
```

```
class medicalAttendance extends Attendance {  
    @Override  
    double calculateAttendance(int attended, int total) {  
        return attended * 100.0 / total;  
    }  
}
```

```
public class main {  
    public static void main(String[] args) {  
        Attendance e = new EngineeringAttendance();  
        System.out.println(  
            "Attendance = " +  
            e.calculateAttendance(45, 50) + "%"  
        );  
    }  
}
```


main.java

Version Control

Current File

Trial

Project

main.java

.idea

out

src

Main.java

.gitignore

main.java

main.java.iml

External Libraries

Scratches and Consoles

Main.java

Trial

```
14 class MedicalAttendance extends Attendance { no usages
15     @Override 1 usage
16     double calculateAttendance(int attended, int total) {
17         return attended * 100.0 / total;
18     }
19 }
20
21 public class Main {
22     public static void main(String[] args) {
23         Attendance e = new EngineeringAttendance();
24
25         System.out.println(
26             "Attendance = " +
27             e.calculateAttendance(attended: 45, total: 50) + "%"
28         );
29     }
30 }
```

Run

Main

C:\Users\acer\.jdk\openjdk-26.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA\lib\idea_rt.jar=12174:C:\Program Files\JetBrains\IntelliJ IDEA\bin\idea64.exe" -classpath C:\Users\acer\IdeaProjects\main\out\production\main Main

Attendance = 90.0%

Process finished with exit code 0

Performance

Show Results

00:00

Process Terminated

main.java

src

Main.java

30:2

LF

UTF-8

4 spaces

25°C

Partly sunny

Search

9:12 AM

6/22/2026

8.) Notification Interface

`</> java`

`Interface Notification {`

`void sendNotification (String message);`

`}`

`class EmailNotification implements Notification {`

`public void sendNotification (String message) {`

`System.out.println ("Email sent: " + message);`

`}`

`}`

`class SMSNotification implements Notification {`

`public void sendNotification (String message) {`

`System.out.println ("SMS sent: " + message);`

`}`

`}`

`public class Main {`

`public static void main (String [] args) {`

`Notification email = new EmailNotification();`

`Notification sms = new SMSNotification();`

`email.sendNotification ("Admission confirmed");`

`sms.sendNotification ("Fee Due");`

`}`

`}`

9.) unique student ID
// Java

```
class student {
    private static int counter = 1000;
    private int id;
```

```
    student() {
        id = ++counter;
```

```
    }
```

```
    void display ID() {
```

```
        System.out.println("student ID:" + id);
```

```
    }
```

```
}
```

```
public class main {
```

```
    public static void main(String[] args) {
```

```
        student s1 = new student();
```

```
        student s2 = new student();
```

```
        s1.display ID();
```

```
        s2.display ID();
```

```
    }
```

```
}
```


main.java

Version Control

Current File

Trial

Project

main.java

.idea

out

src

Main.java

.gitignore

main.java

main.java.iml

External Libraries

Scratches and Consoles

Main.java

Trial

```
1  class Student { 4 usages
2      id = 1000000;
3
4
5
6
7      }
8
9      void displayID() { 2 usages
10         System.out.println("Student ID: " + id);
11     }
12 }
13
14 public class Main {
15     public static void main(String[] args) {
16         Student s1 = new Student();
17         Student s2 = new Student();
18
19         s1.displayID();
20         s2.displayID();
21     }
22 }
```

Run

Main

C:\Users\acer\.jdk\openjdk-26.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA\bin\idea_rt.jar=12174:C:\Program Files\JetBrains\IntelliJ IDEA\bin\idea.jar" -Didea.launcher.debug=off

Student ID: 1001
Student ID: 1002

Process finished with exit code 0

Performance

Show Results

00:00

Process Terminated

main.java

src

Main.java

22:2

LF

UTF-8

4 spaces

25°C

Partly sunny

Search

9:13 AM

6/22/2026

20.) scholarship
// Java

```
abstract class scholarship {  
    abstract boolean isEligible (double value);  
}
```

```
class merit scholarship extends scholarship {  
    boolean isEligible (double percentage) {  
        return percentage >= 80;  
    }  
}
```

```
class Need Based scholarship extends scholarship {  
    boolean isEligible (double income) {  
        return income <= 200000;  
    }  
}
```

```
{  
    public class main {  
        public static void main (String [] args) {  
            scholarship merit = new merit scholarship ();  
            scholarship merit = new Need Based scholarship ();  
  
            system.out.println ("merit Eligible: " + merit.  
                                isEligible (85));  
            system.out.println ("Need Based Eligible: "  
                                + merit.isEligible (250000));  
        }  
    }  
}
```


